

Example:

Input (3-variables)			Output (Y)
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

∴ Notation of SOP expression is:

$$f(A, B, C) = \sum m(3, 5, 6, 7)$$

$$Y = m_3 + m_5 + m_6 + m_7$$

$$\text{Also, } Y = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

2.5.2. POS Form

The POS expression usually takes the form of two or more OR variables within parentheses, ANDed with two or more such terms.

Example: $Y = (A + \bar{B} + C)(\bar{B}\bar{C} + D)$

Each individual term in standard POS form is called maxterm.

POS forms are used to write logical expression for output be coming logic '0'.

we get $f(A, B, C) = \pi M(0, 1, 2, 4)$

$$Y = M_0.M_1.M_2.M_4$$

$$Y = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C)$$

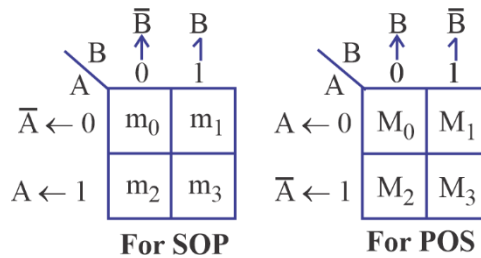
∴ We can also conclude from Table 2 and from above equations:

if $Y = \sum m(3, 5, 6, 7)$ or $Y = \pi M(0, 1, 2, 4)$

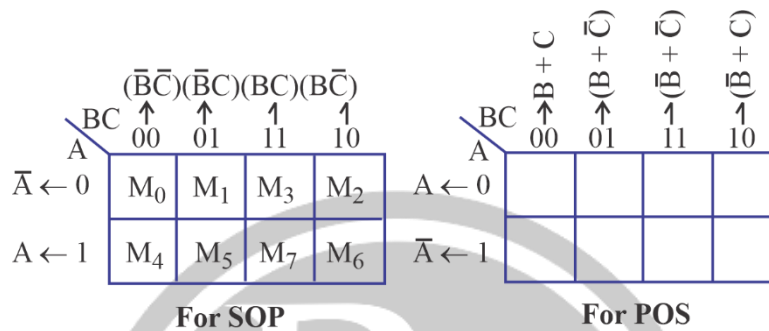
2.6. Karnaugh Map (K-MAP)

The K-map is a graphical method which provides a systematic method for simplifying the Boolean expressions. In n variable K-map, there are 2^n cells.

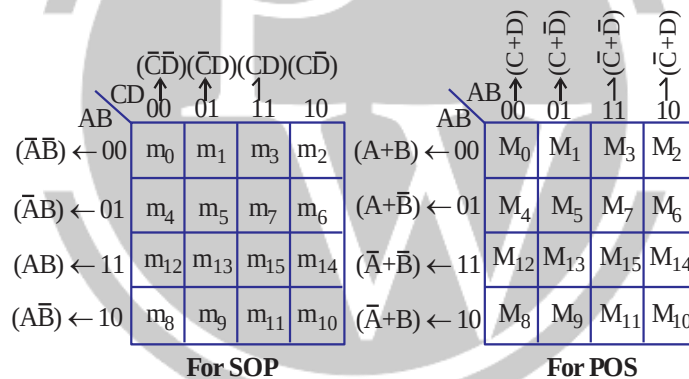
2.6.1. Two variable K-map



2.6.2. Three variable K-map

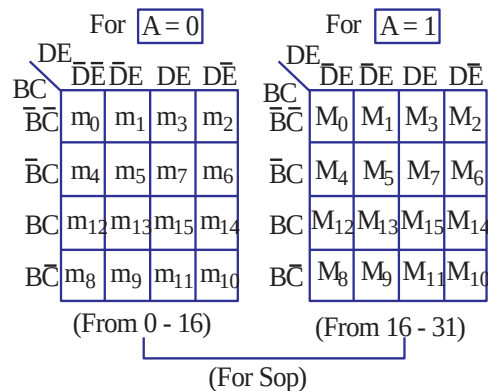


2.6.3. Four Variable K-map



2.6.4. Five variable K-map

- 32 cells
 - 32 Minterms (Maxterms)
- Here, we have $f(A, B, C, D, E)$



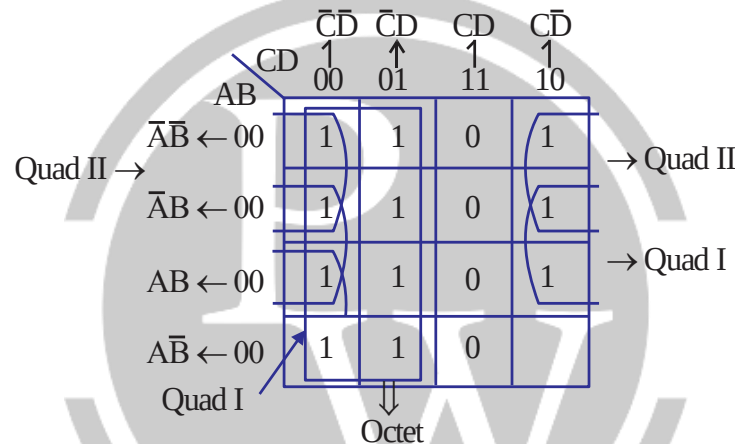
2.7. Simplification Rules

1. Construct the K-map and place 1's in those cells corresponding to the 1's in the truth table. Place the 0's in the other cells.
2. Examine the map for adjacent 1's and loop those 1's which are not adjacent to any other 1's. These are called isolated 1's.
3. Next, look for those 1's which are adjacent to only one other 1. Loop any pair containing such a 1.
4. Loop any octet even if it contains some 1's that have already been looped.
5. Loop any quad that contains one or more 1's which have not already been looped, making sure to use the minimum number of loops.
6. Form the OR sum of all the terms generated by each loop.

Example: Simplify a four variable logic function using K-map

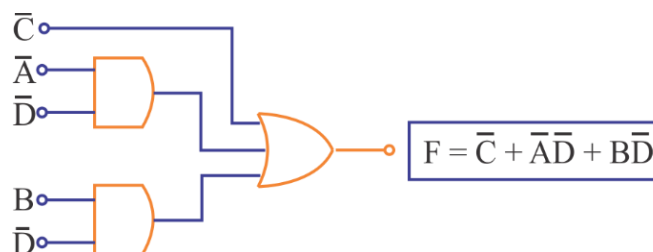
$f(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$ also implement the simplified expression with AND-OR logic.

Solution:



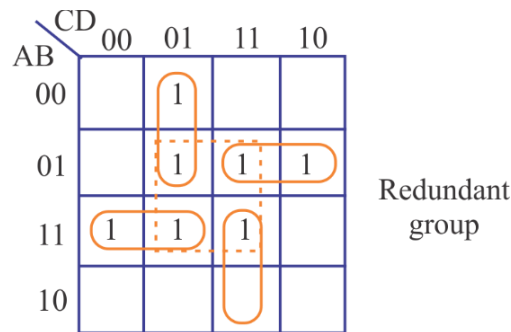
$$\begin{aligned}
 f &= \bar{C} + \bar{A}\bar{D} + B\bar{D} \\
 \therefore &\quad \downarrow \quad \downarrow \quad \downarrow \\
 &\text{octet} \quad \text{quad - II} \quad \text{quad - I}
 \end{aligned}$$

⇒ Gate implementation:



2.8. Redundant Group

If all the 1's in a group are already involved in some other groups, then that group is caused as a redundant group. A redundant group has to be eliminated, because it increases the no of gates required.



2.9. Don't Care Condition

The combinations for which the values of the expression are not specified are called don't care conditions.

Example: Simply the given equation in part (i) and (ii)

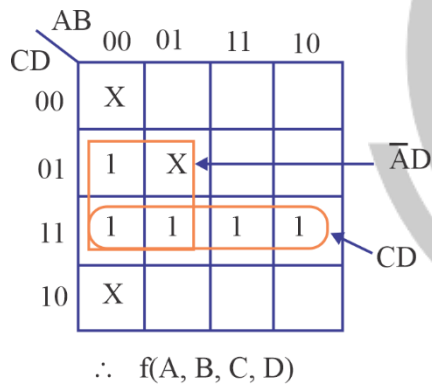
(i) In terms of SOP. and don't care conditions

$$f(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + d(0, 2, 5)$$

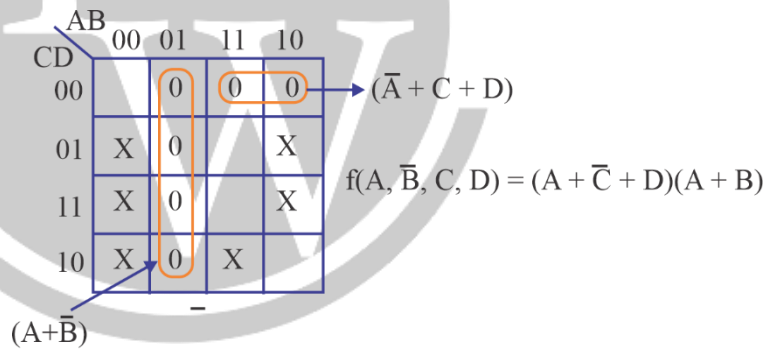
(ii) In terms of POS and don't care conditions.

$$f(A, B, C, D) = \pi M(4, 5, 6, 7, 8, 12) + d(1, 2, 3, 9, 11, 14)$$

Solution:



(i)



(ii)

2.10. Implicants, Prime Implicants and Essential Prime Implicants

2.10.1. Implicants

Implicants is a product term on the given function for that combination the function output must be 1.

2.10.2. Prime Implicant (PI)

Prime implicant is a smallest possible product term of the given function,

2.10.3. Essential Prime Implicants (EPI)

EPI is a prime implicant it must cover at least one minterms, which is not covered by other PI.

Example: Reduce the expression using mapping $F = \Sigma m(2, 3, 6, 7, 8, 10, 11, 13, 14)$

Solution :

CD \ AB	00	01	11	10
00			1	1
01			1	1
11		1		1
10	1		1	1

$$F = ABCD + ABD + \bar{A}C + \bar{B}C + C\bar{D}$$

Example : Reduce the following expression using k-map and identify PI's and EPI

$$F = \Sigma m(0, 1, 2, 3, 6, 7, 13, 15)$$

Solution :

CD \ AB	00	01	11	10
00	1	1	1	1
01			1	1
11			1	1
10				

$$EPI = \bar{A}\bar{B}, \bar{A}C, ABD$$

$$PI = BCD$$

$$\text{Minimal } F = \bar{A}\bar{B} + \bar{A}C + ABD$$

Example: Given the following Karnaugh map, which one of the following represents the minimal sum of products of the map.

- A. $xy + \bar{y}z$
- B. $\omega \bar{x} \bar{y} + xy + xz$
- C. $\bar{\omega}x + \bar{y}z + xy$
- D. $xy + y$

Solution :

$\omega x \backslash yz$	00	01	11	10
00	0	x	0	x
01	x	x	x	1
11	0	x	1	0
10	0	1	x	0

$\omega x \backslash yz$	00	01	11	10
00	0	x	0	x
01	x	x	x	1
11	0	x	1	0
10	0	1	x	0

$$= xy + \bar{y}z$$

